

How to Go About Linux Boot Time Optimisation



Booting an embedded device or a piece of telecommunication equipment quickly is crucial for time-critical applications and also plays a very major role in improving the user experience. This article gives some important tips on how to enhance the boot-up time of any device.

Fast booting or fast rebooting plays a crucial role in various situations. It is critical for an embedded system to boot up fast in order to maintain the high availability and better performance of all the services. Imagine a telecommunications device running a Linux operating system that does not have fast booting enabled. All the systems, services and the users dependent on that particular embedded device might be affected. It is really important that devices maintain high availability in their services, for which fast booting and rebooting play a crucial role.

A small failure or shutdown of a telecom device, even for a few seconds, can play havoc with countless users working on the Internet. Thus, it is really important for a lot of time-dependent devices and telecommunication devices to incorporate fast booting in their devices to help them get back to work quicker. Let us understand the Linux boot-up procedure from Figure 1.

Monitoring tools and the boot-up procedure

A user should take note of a number of factors before making changes to a machine. These include the current booting speed of the machine and also the services, processes or applications that are taking up resources and increasing the boot-up time.

Boot chart: To monitor the boot-up speed and the various services that start while booting up, the user can install the

boot chart using the following command:

```
sudo apt-get install pybootchartgui.
```

Each time you boot up, the boot chart saves a *.png* (portable network graphics) file in the log, which enables the user to view the *.png* files to get an understanding about the system's boot-up process and services. Use the following command for this purpose:

```
cd /var/log/bootchart
```

The user might need an application to view the *.png* files. Feh is an X11 image viewer that targets console users. It doesn't have a fancy GUI, unlike most other image viewers, but it simply displays pictures. Feh can be used to view the *.png* files. You can install it using the following command:

```
sudo apt-get install feh
```

You can view the *.png* files using *feh xxxx.png*.

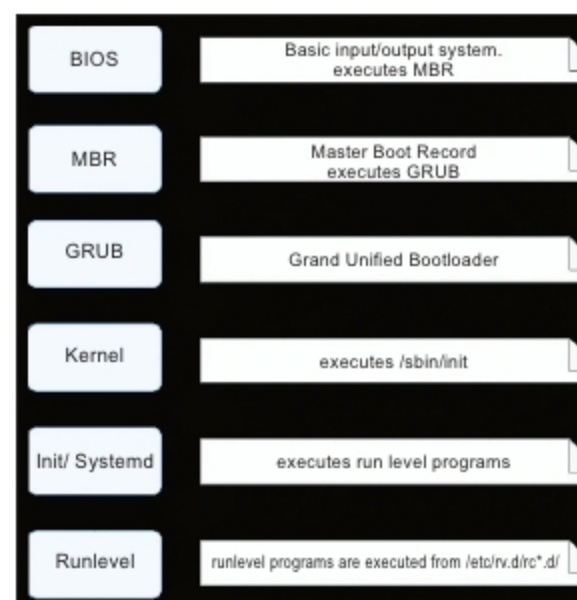


Figure 1: Boot-up procedure